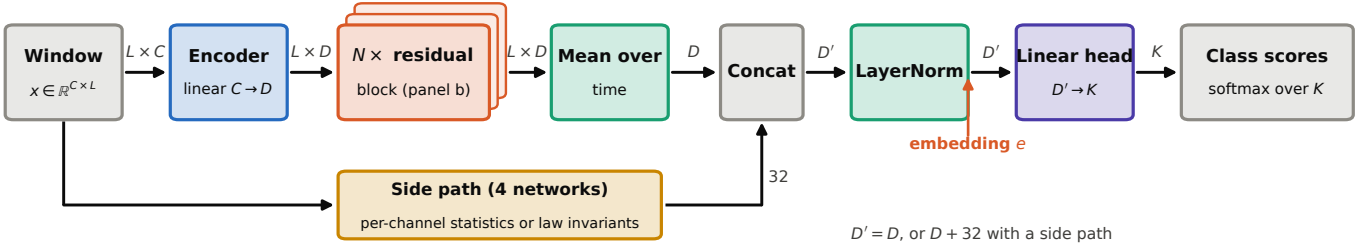
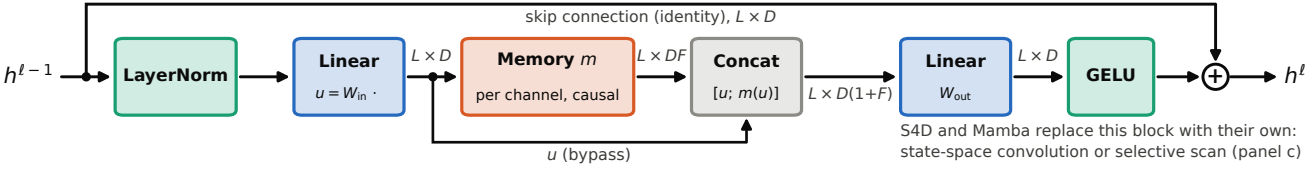
(a) Shared skeleton of the ten networks



(b) Residual block $\ell = 1, \ldots, N$ (recurrence, play, quantile and plain blocks)



(c) Memories: update per channel and time step t (input u_t)

Memory	Update rule	State per channel	Networks
Recurrence plot	$r_t(\tau) = \sigma((\varepsilon - \lVert z_t - z_{t-\tau} \rVert)/T)$ and $r_t(\tau)\, r_{t-1}(\tau)$, z_t: 8 learned 2-D projections of u_t; 10 lags $\tau \le 48$	48-sample buffer	RecurrenceNet, RN + stats, RN + laws
Play operator	$y_t = \min(\max(y_{t-1},\, u_t - r),\, u_t + r)$, 4 widths r (HysteresisNet: r depends on u_t, plus a clock channel)	4 values y	HysteresisNet; No memory ($r = 0$)
Quantile tracker	$q_t = q_{t-1} + \eta\,(\tau - \mathbf{1}[u_t < q_{t-1}])$, 4 levels τ	4 values q	QuantileNet
Diagonal SSM	$s_t = \bar{A}\, s_{t-1} + \bar{B}\, u_t$, $y_t = \mathrm{Re}(C\, s_t) + D u_t$; FFT convolution	16 complex modes	S4D
Selective scan	$s_t = \bar{A}_t\, s_{t-1} + \bar{B}_t\, u_t$, $y_t = C_t\, s_t$; Δ_t, B_t, C_t computed from u_t	16 states	Mamba
None	plain block: the memory and the concatenation are absent	none	Laws, Stats (no mem.)